ASSIGNMENT 2

SMART AGRICULTURE MONITORING AND AUTOMATED IRRIGATION USING ESP8266 AND BLYNK

1. Introduction

Agriculture plays an important role in food production and economic development. One of the major challenges in agriculture is maintaining an appropriate moisture level in the soil. Manual monitoring of soil conditions can be time-consuming and may result in either over-irrigation or under-irrigation.

To demonstrate an IoT-based solution to this problem, a **Smart Agriculture Monitoring and Automated Irrigation System** was developed using an **ESP8266, slide potentiometer, relay module, bipolar DC motor, and Blynk IoT platform**.

Since this project is implemented as a prototype/simulation, a **slide potentiometer is used to simulate the variable output of a soil moisture sensor**. By changing the potentiometer position, different soil-moisture conditions can be represented. The ESP8266 reads the analog value from the potentiometer and processes it as the soil moisture level.

A **bipolar DC motor is used to represent the irrigation pump**. The motor is controlled through a relay according to the soil-moisture condition.

The sensor-simulation value is transmitted to the Blynk IoT platform through Wi-Fi, allowing the user to monitor the soil condition remotely.

2. Objectives

The main objectives of this assignment are:

- To understand the application of IoT in smart agriculture.
- To simulate soil-moisture variation using a slide potentiometer.
- To interface an analog input with the ESP8266.
- To transmit sensor-related data to the Blynk IoT platform.
- To remotely monitor the simulated soil-moisture condition.
- To control a relay-based irrigation system.
- To use a bipolar DC motor as a simulated irrigation pump.
- To understand analog sensor data processing.
- To demonstrate automatic irrigation logic using predefined thresholds.
- To understand the application of IoT in agricultural automation.

3. Problem Statement

Traditional irrigation systems often depend on manual observation and fixed irrigation schedules. Such methods may result in excessive watering or insufficient watering.

The objective of this project is to develop a prototype IoT-based agricultural monitoring system in which the soil-moisture condition can be simulated, monitored remotely, and used to control an irrigation load.

The system should:

1. Generate a variable analog value representing soil moisture.
2. Read the analog value using the ESP8266.
3. Process the input according to predefined moisture conditions.
4. Connect the ESP8266 to the Internet through Wi-Fi.
5. Send the moisture-related value to the Blynk platform.
6. Display the value on the Blynk dashboard.
7. Control the relay according to the simulated soil condition.
8. Operate the bipolar DC motor when irrigation is required.

4. Proposed Solution

The proposed system uses an **ESP8266** as the main controller.

A **slide potentiometer** is connected to an analog input of the ESP8266. The potentiometer provides a variable analog voltage that is used to simulate different soil-moisture conditions.

The ESP8266 reads this analog value and determines whether the simulated soil condition is sufficiently moist or dry according to the programmed threshold.

The sensor-related value is also transmitted to the **Blynk IoT platform** through Wi-Fi, where it can be monitored using a dashboard.

A **relay module** is used to control a **bipolar DC motor**, which represents the irrigation pump.

The overall system can be represented as:

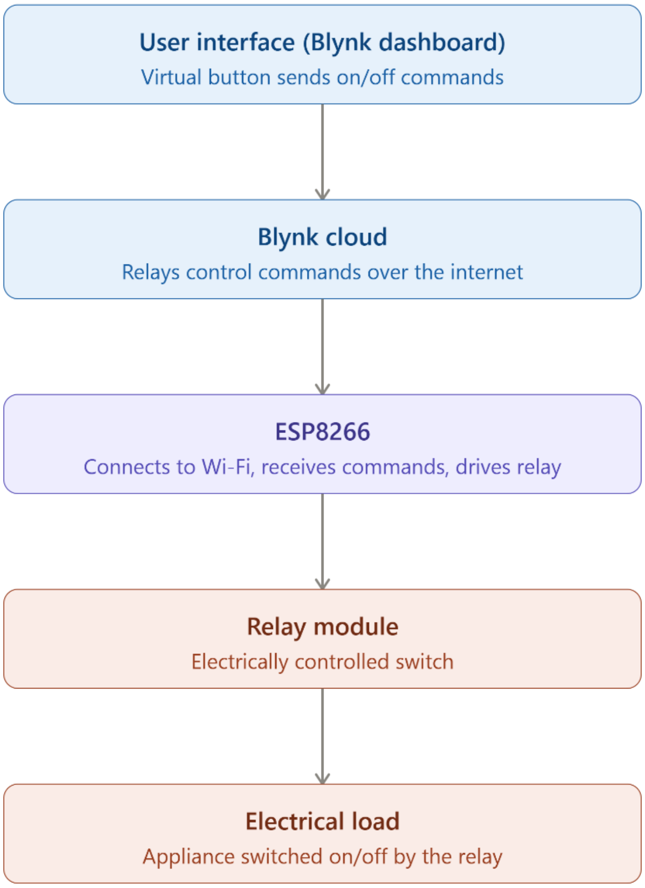
Slide Potentiometer → ESP8266 → Wi-Fi → Blynk Cloud → Dashboard

For irrigation control:

ESP8266 → Relay → Bipolar DC Motor

5. System Architecture

The system consists of the following major blocks:



Low-voltage control signal (ESP8266 → relay) switches the higher-voltage load

5.1 Slide Potentiometer

The slide potentiometer provides a variable analog output. In this prototype, it is used to simulate changes in soil moisture.

Moving the potentiometer changes its output voltage, which produces different analog readings at the ESP8266.

5.2 ESP8266

The ESP8266 acts as the central controller. It reads the analog value, processes the input, communicates with Blynk, and controls the relay.

5.3 Wi-Fi Network

The ESP8266 uses its built-in Wi-Fi capability to establish Internet connectivity.

5.4 Blynk Cloud

Blynk Cloud provides communication between the ESP8266 and the user dashboard.

5.5 Blynk Dashboard

The Blynk dashboard displays the simulated soil-moisture reading and provides a remote interface for monitoring and controlling the system.

5.6 Relay Module

The relay acts as an electronic switch between the ESP8266 and the bipolar DC motor.

5.7 Bipolar DC Motor

The bipolar DC motor is used as a **simulation/representation of an irrigation pump**. When the relay is activated, the motor operates to represent the irrigation process.

6. Hardware Requirements

Sl. No.	Component	Purpose
1	ESP8266	Main controller and Wi-Fi communication
2	Slide Potentiometer	Simulates variable soil-moisture conditions
3	Relay Module	Controls the irrigation load
4	Bipolar DC Motor	Represents the irrigation pump
5	Power Supply	Provides power to the circuit
6	Connecting Wires	Electrical connections
7	Breadboard / Simulation Platform	Circuit implementation

7. Software Requirements

Software/Platform	Purpose
Arduino IDE	ESP8266 programming
Embedded C/C++	Firmware development
Blynk IoT	IoT communication and monitoring
Wi-Fi	Wireless Internet connectivity
Blynk Dashboard	Remote monitoring and control

8. ESP8266

The ESP8266 is a Wi-Fi-enabled microcontroller widely used in IoT applications.

In this project, the ESP8266 performs the following functions:

- Reads the analog output of the slide potentiometer.
- Processes the analog value as a simulated soil-moisture reading.
- Determines the simulated soil condition.
- Connects to the Wi-Fi network.
- Communicates with Blynk Cloud.
- Sends the analog reading to the dashboard.
- Controls the relay according to the programmed irrigation logic.

The built-in Wi-Fi capability of the ESP8266 makes it suitable for IoT-based agricultural prototypes.

9. Slide Potentiometer as Soil-Moisture Simulator

A slide potentiometer is a variable resistor that produces a variable voltage when connected appropriately between the supply and ground.

In this project, **the slide potentiometer is used instead of a physical soil-moisture sensor.**

The position of the potentiometer is manually changed to simulate different soil conditions.

For example:

- One end of the potentiometer range can represent **dry soil**.
- A middle position can represent **moderate soil moisture**.
- The opposite end can represent **high soil moisture**.

The ESP8266 reads the potentiometer's analog output and uses the reading as the simulated soil-moisture value.

This approach is useful for testing the control logic without requiring a physical soil-moisture sensor.

10. Relay Module

The relay module is used to control the bipolar DC motor.

The ESP8266 GPIO provides the control signal to the relay. When the programmed condition indicates that irrigation is required, the relay changes its state and supplies power to the motor.

The relay therefore provides an interface between the low-power ESP8266 control circuit and the motor load.

11. Bipolar DC Motor as Irrigation Pump

A bipolar DC motor is used in the prototype to represent the operation of an irrigation pump.

The motor does not physically pump water in this implementation. Instead, its ON/OFF operation represents the operation of a water pump.

The control logic is:

Dry condition → Relay ON → Motor ON → Irrigation represented

Moist condition → Relay OFF → Motor OFF → Irrigation stopped

This provides a simple visual and functional representation of automatic irrigation.

12. Blynk IoT Platform

Blynk is used as the IoT platform for remote monitoring and control.

The Blynk dashboard allows the user to:

- View the simulated soil-moisture value.
- Monitor the changing analog reading.
- Observe the irrigation/motor status.
- Control or monitor the relay state.
- Receive real-time information from the ESP8266.

The Blynk platform provides a convenient interface between the user and the agricultural prototype.

13. Analog Value and Soil-Moisture Simulation

The slide potentiometer generates an analog value that is read by the ESP8266.

The ESP8266 processes this value according to the threshold configured in the program.

The analog value is used to represent different soil conditions.

Simulated Condition	Potentiometer / Analog Reading	System Response
Dry	Value represents low moisture	Relay ON, Motor ON
Moderate	Intermediate value	System monitors condition

Simulated Condition	Potentiometer / Analog Reading	System Response
Wet / Sufficient Moisture	Value represents sufficient moisture	Relay OFF, Motor OFF

The exact threshold depends on the values configured in the program.

Since the potentiometer is a simulation of the sensor rather than an actual soil-moisture sensor, the values represent **simulated soil conditions**.

14. Working Principle

The working of the system can be explained as follows:

Step 1: Power ON

The ESP8266, slide potentiometer, relay module, and motor circuit are powered.

Step 2: Simulated Soil Condition

The position of the slide potentiometer is changed to represent a particular soil-moisture condition.

Step 3: Analog Data Acquisition

The ESP8266 reads the potentiometer's analog output.

Step 4: Data Processing

The ESP8266 compares the analog value with the predefined threshold.

Step 5: Blynk Communication

The ESP8266 sends the analog reading to Blynk Cloud through Wi-Fi.

Step 6: Dashboard Display

The simulated soil-moisture reading is displayed on the Blynk dashboard.

Step 7: Irrigation Decision

If the programmed condition indicates that the soil is dry, the ESP8266 activates the relay.

Step 8: Motor Activation

The relay switches ON the bipolar DC motor, representing the operation of the irrigation pump.

Step 9: Irrigation Stop

When the simulated soil condition reaches the required moisture level, the ESP8266 switches OFF the relay and the motor stops.

15. Algorithm

Algorithm for Smart Agriculture Monitoring and Irrigation

Start

↓

Initialize ESP8266

↓

Initialize slide potentiometer analog input

↓

Initialize relay

↓

Connect to Wi-Fi

↓

Connect to Blynk Cloud

↓

Read analog value from slide potentiometer

↓

Process the analog value

↓

Send value to Blynk

↓

Display value on dashboard

↓

Compare value with threshold

↓

If simulated soil condition = Dry

→ Activate relay

→ Turn ON bipolar DC motor

→ Represent irrigation ON

↓

If simulated soil condition = Moist

→ Deactivate relay

→ Turn OFF bipolar DC motor

→ Represent irrigation OFF

↓

Repeat monitoring

↓

Continue operation

16. Software Implementation

The firmware was developed using the Arduino IDE.

The main software operations include:

1. Including the required libraries.
2. Defining Wi-Fi credentials.
3. Configuring Blynk authentication.
4. Defining the analog input pin.
5. Defining the relay output pin.
6. Initializing the analog input.
7. Initializing the relay.
8. Connecting the ESP8266 to Wi-Fi.
9. Establishing communication with Blynk.
10. Reading the slide potentiometer value.
11. Processing the analog value.
12. Sending the value to Blynk.
13. Updating the dashboard.
14. Comparing the value with the configured threshold.
15. Activating or deactivating the relay.
16. Controlling the bipolar DC motor accordingly.

17. Dashboard Implementation

A Blynk dashboard was developed to monitor the smart agriculture prototype.

The dashboard provides information related to:

- Simulated soil-moisture value.
- Relay status.
- Motor/irrigation status.
- Analog sensor reading.

The dashboard allows the user to observe changes in the simulated soil condition remotely.

The dashboard screenshots are maintained separately as part of the assignment evidence.

18. Testing

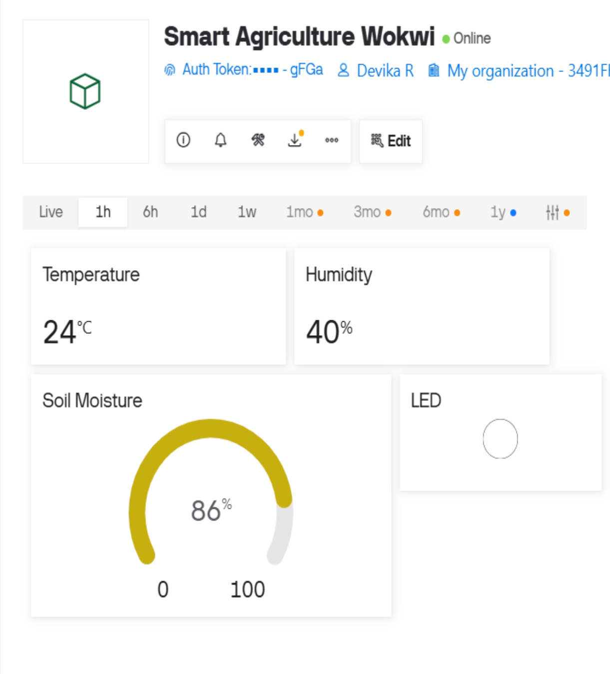
The system was tested by changing the position of the slide potentiometer to represent different soil conditions.

Test Case	Input / Condition	Expected Result	Observed Result	Status
1	ESP8266 powered ON	Controller starts	Controller started	Pass
2	Wi-Fi available	ESP8266 connects	Connected	Pass
3	Blynk available	Dashboard communication established	Communication established	Pass
4	Potentiometer value changed	Analog reading changes	Reading changed	Pass
5	Dry condition simulated	Relay should activate	Relay activated	Pass
6	Dry condition simulated	Motor should turn ON	Motor turned ON	Pass
7	Moist condition simulated	Relay should deactivate	Relay deactivated	Pass
8	Moist condition simulated	Motor should turn OFF	Motor turned OFF	Pass
9	Dashboard monitoring	Analog value should be displayed	Value displayed	Pass

19. Test Evidence

The screenshots captured during testing provide evidence of the system operation.

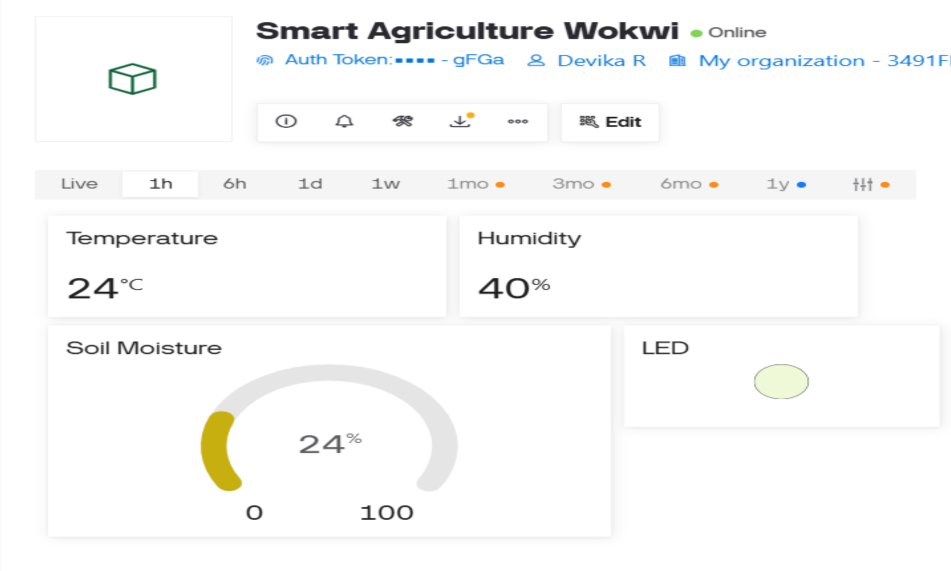
19.1 Blynk Dashboard – Light OFF / Relay OFF



blynk_dashboard_lightoff.png

This screenshot shows the Blynk dashboard when the relay-controlled load is in the OFF condition.

19.2 Blynk Dashboard – Light ON / Relay ON



blynk_dashboard_lighton.png

This screenshot shows the Blynk dashboard when the relay-controlled load is in the ON condition.

19.5 Serial Monitor

```
Soil Status : DRY
Pump Status : ON
----------------------------------------
Temperature : 24.00 °C
Humidity    : 40.00 %
Soil Raw    : 152
Soil Moisture : 3 %
```

serial_monitor.png

The Serial Monitor was used to observe the ESP8266 operation, analog readings, and communication status.

20. Results

The Smart Agriculture Monitoring and Automated Irrigation prototype was successfully implemented and tested.

The slide potentiometer successfully generated variable analog values that were used to simulate different soil-moisture conditions. The ESP8266 successfully read and processed these values.

The processed values were transmitted to the Blynk IoT platform through Wi-Fi and displayed on the dashboard.

The relay was successfully controlled according to the programmed threshold, and the bipolar DC motor responded accordingly.

The testing demonstrated:

- Successful ESP8266 operation.
- Successful Wi-Fi connectivity.
- Successful Blynk communication.
- Successful analog input reading.
- Successful simulation of soil-moisture variation.
- Successful dashboard monitoring.
- Successful relay operation.
- Successful motor control.
- Successful hardware/software integration.

Therefore, the prototype successfully demonstrated the basic operation of an IoT-based smart agriculture and automated irrigation system.

21. Advantages

The proposed prototype provides several advantages:

- Simple and low-cost implementation.
- Easy simulation of changing soil conditions.
- Real-time monitoring through Blynk.
- Remote monitoring using Wi-Fi.
- Demonstrates automatic irrigation logic.
- Easy to test different threshold conditions.
- Provides a foundation for integrating a real soil-moisture sensor.
- Can be expanded with additional agricultural sensors.

22. Limitations

The prototype has the following limitations:

- The slide potentiometer does not physically measure soil moisture.
- The potentiometer only simulates the output of a soil-moisture sensor.
- The bipolar DC motor represents the irrigation pump but does not perform actual water pumping in the prototype.
- The system depends on Wi-Fi and Internet connectivity for remote Blynk monitoring.
- A real agricultural implementation would require sensor calibration.
- A practical irrigation system would require a suitable water pump and appropriate power circuitry.
- Outdoor deployment would require weatherproof hardware.

23. Future Scope

The prototype can be further developed by replacing the simulated components with actual agricultural hardware.

23.1 Real Soil Moisture Sensor

The slide potentiometer can be replaced with a capacitive or suitable soil-moisture sensor to obtain actual soil moisture measurements.

23.2 Real Water Pump

The bipolar DC motor can be replaced with a suitable water pump for actual irrigation.

23.3 Multiple Sensors

Additional sensors can be incorporated, including:

- Temperature sensor
- Humidity sensor
- pH sensor
- NPK sensor
- Light sensor
- Water-level sensor

23.4 Weather Integration

Weather information and rainfall prediction can be integrated to avoid unnecessary irrigation.

23.5 Data Logging

Historical sensor data can be stored in the cloud for monitoring and analysis.

23.6 Mobile Notifications

Notifications can be generated when the soil becomes dry or irrigation starts/stops.

23.7 AI-Based Irrigation

Machine learning can be incorporated to predict irrigation requirements based on soil, weather, crop, and historical data.

23.8 Solar-Powered System

Solar energy and battery storage can be added for operation in remote agricultural areas.

24. Applications

The developed prototype can be used as a foundation for:

- Smart agriculture systems.
- Automated irrigation systems.
- Greenhouse automation.
- Garden monitoring.
- Nursery automation.
- Terrace garden systems.
- IoT-based crop monitoring.
- Agricultural research prototypes.
- Educational IoT projects.

25. Learning Outcomes

Through this assignment, the following concepts and skills were developed:

- Understanding of IoT-based smart agriculture.
- ESP8266 programming.
- Embedded C/C++ programming.
- Analog input interfacing.
- Variable analog signal generation using a potentiometer.
- Relay interfacing.
- DC motor control.
- Wi-Fi communication.
- Blynk IoT platform configuration.
- Dashboard development.
- Threshold-based control logic.
- Hardware-software integration.
- Debugging using Serial Monitor.
- Functional testing of an IoT prototype.

The assignment provided practical experience in developing an IoT system in which an analog input is processed by a microcontroller, transmitted to a cloud platform, and used to control an output device.

26. Conclusion

The **Smart Agriculture Monitoring and Automated Irrigation System using ESP8266 and Blynk** was successfully designed, implemented, and tested as an IoT prototype.

In this implementation, a **slide potentiometer was used to simulate soil-moisture variations**, allowing different moisture conditions to be generated and tested. The ESP8266 read the analog value, processed the input according to the programmed threshold, and transmitted the information to the Blynk IoT platform.

A **bipolar DC motor was used to represent the irrigation pump**, with a relay controlling its ON/OFF operation according to the simulated soil condition.

The project successfully demonstrates the fundamental concept of **IoT-based agricultural monitoring and automated irrigation**. The prototype can be further improved by replacing the simulated potentiometer with a real soil-moisture sensor and the DC motor with an actual irrigation pump.

27. References

1. ESP8266 technical documentation and reference materials.
2. Arduino IDE documentation.
3. Blynk IoT documentation.
4. Potentiometer technical/reference documentation.
5. Relay module technical documentation.
6. DC motor technical/reference documentation.
7. IoT and smart agriculture learning resources used during implementation.

28. Appendix

A. Project Files

The project repository contains the source code, README file, documentation, and testing evidence.

Important screenshot files include:

- blynk_dashboard_lightoff.png
- blynk_dashboard_lighton.png
- relay_lightoff.png
- relay_lighton.png
- serial_monitor.png
- soil_moisture.png

B. Source Code

The complete ESP8266 source code used for the smart agriculture prototype is maintained in the project repository.

C. Screenshots

The screenshots are maintained separately as part of the assignment submission. They provide visual evidence of the Blynk dashboard, relay operation, Serial Monitor output, and simulated soil-moisture readings.

D. Wokwi Project: